

VAISHNAV A

AI/ML Engineer | Computer Vision & Deep Learning | Full-Stack Engineer (React.js, Flutter, Python)

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SUMMARY

AI/ML-focused Software Engineer with 3+ years of experience across the full SDLC, combining hands-on deep learning work with production web and mobile development. Designed and trained a Vision Transformer (ViT) for multi-class medical image classification (chest X-ray screening for pneumonia, tuberculosis, and COVID-19), plus CNN transfer-learning models (ResNet, EfficientNet) and YOLO/MediaPipe computer-vision prototypes, using PyTorch, TensorFlow, and OpenCV. Paired this with full-stack engineering — Python/Django and Flask/MySQL backends, IoT sensor pipelines (ESP32, MAX30102, GSR), and React.js/Flutter applications serving 1,000+ active users. Strong foundation in data structures, OOP, and RESTful API development, with hands-on Firebase, SQL/NoSQL, Git, and Agile/Scrum experience.

KEY SKILLS

Applied AI/ML: Computer Vision, Vision Transformer (ViT) Architecture, Hugging Face Transformers, CNNs & Transfer Learning (ResNet, EfficientNet), Object Detection (YOLO), MediaPipe (Hand/Gesture Tracking), Model Fine-Tuning, Image Classification, Data Preprocessing, IoT Sensor Integration (ESP32, MAX30102, GSR), Health/Signal Data Processing

AI/ML Libraries & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

Core Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), Design Patterns, SDLC, Agile/Scrum, Unit Testing, Debugging, Code Reviews

Primary Stack (Professional Experience): React.js, Flutter (Dart), Python, Django, JavaScript (ES6+), HTML5, CSS3, Redux, Context API, React Hooks, REST API Integration, Firebase (Auth, Firestore, Cloud Storage), Celery, Git & GitHub, Postman

Tools & Practices: CI/CD Basics, Agile/Scrum, Cross-Browser Development, Performance Optimization, Accessibility (WCAG), Google Maps, Payment Gateways (Razorpay), Push Notifications, WebSockets, IoT

Exploratory / Self-Directed: LLM & AI API Integration Basics — applied in personal projects

EXPERIENCE

Software Engineer — Trycode Innovations

Sep 2024 – Present | Kozhikode, Kerala

- Designed, built, and trained a Vision Transformer (ViT)-based deep learning model for multi-class chest X-ray classification (pneumonia, tuberculosis, COVID-19) as part of PulmoGuard AI, an IoT-integrated patient monitoring system; built the supporting Flask/MySQL backend and health-scoring engine processing real-time SpO₂, heart-rate, and GSR sensor data from an ESP32-based wearable device.
- Built prototype CNN-based image classifiers (ResNet/EfficientNet via transfer learning) for additional medical imaging use cases — skin disease, brain tumor, and eye disease detection — trained on Kaggle datasets as proof-of-concept work exploring AI-assisted diagnostics beyond the PulmoGuard platform.
- Built additional computer-vision prototypes using YOLO for object detection (fire detection, crowd density/counting) and MediaPipe + CNN for sign language recognition from hand gestures — trained and evaluated on Kaggle/public datasets using PyTorch/TensorFlow and OpenCV for video frame preprocessing.
- Shipped 15+ production features across 3 React.js/Flutter applications in 2-week Agile/Scrum sprints, partnering with a cross-functional team of 5–6 engineers and designers.
- Accelerated front-end load time by 35% and cut new-feature build time by ~20% by architecting code-splitting, lazy loading, memoization, and a shared reusable component library.
- Engineered 10+ RESTful API endpoints integrated with Firebase for real-time sync, authentication, and cloud storage, sustaining 1,000+ active users with near-zero downtime.
- Slashed production bugs by ~25% through rigorous code reviews and unit testing; built a Python/scikit-learn proof-of-concept recommendation feature evaluated by senior engineers.

Freelance Software Engineer — WayToFresh

Mar 2026 – Present | Kozhikode, Kerala (Remote)

- Architecting and building WayToFresh, a full-stack delivery platform, as sole developer — a Flutter customer app, a Flutter delivery-partner app, and a Django REST admin dashboard.
- Implementing real-time order tracking via WebSockets, live rider tracking with Google Maps API, Firebase Cloud Messaging push notifications, OTP authentication, and a full cart, checkout, and order-assignment flow.
- Owning the full stack solo — mobile front-end, backend APIs, and system architecture.

Software Engineer Intern — Nexus Technologies

Sep 2023 – Sep 2024 | Calicut, Kerala

- Built and maintained 2+ responsive web applications in React.js and JavaScript, standardizing UI behavior across screen sizes via reusable component patterns.
- Automated data sync and authentication flows by integrating RESTful APIs with Firebase, cutting manual data-entry steps by ~30%.
- Converted UI/UX wireframes into responsive, accessible front-end interfaces over a 12-month internship, supporting 2 product launches and driving a ~15% reduction in post-release issues.

PROJECTS

Code samples: github.com/Yyshnav

PulmoGuard AI (Vision Transformer, IoT, Flask) (2026) — AI- and IoT-based pulmonary disease prediction and patient monitoring system. Trained a Vision Transformer model on a Kaggle medical imaging dataset to classify chest X-rays across pneumonia, tuberculosis, and COVID-19. Built an ESP32-based wearable unit (MAX30102, GSR sensors) streaming real-time SpO₂ and stress data to a Flask/MySQL backend; generated automated health-score alerts to a companion mobile app for caregivers.

GroceryGo (Flutter & Riverpod) — Built a hyper-local quick-commerce grocery delivery app with 5 km dark-store discovery, sub-15-minute delivery tracking, real-time deal filters, Razorpay payments, and in-store QR self-checkout.

Inco Web Application — Reward & point redemption platform built with React.js and Firebase (real-time data, authentication).

Rawabi Food Delivery & Restaurant — Food delivery platform with a customer-facing website and a delivery-partner web app; integrated Google Maps and Firebase.

Mobile App Prototype (Flutter) — Cross-platform mobile UI built with Flutter/Dart, consuming REST APIs for data and Firebase for authentication.

EDUCATION

Master of Computer Applications — AWH Engineering College, Kuttikkattoor (Dr. A.P.J. Abdul Kalam Technical University)

Jan 2022 – Jun 2024 | Calicut, Kerala

Bachelor of Science, Computer Science — Calicut University, Thenhipalam

Jan 2018 – Oct 2021 | Malappuram, Kerala

CERTIFICATIONS

Software Engineering Internship — Nexus Technologies (2024) | Cloud Computing — NPTEL, IIT Kharagpur (2021) | Learning Git and GitHub — LinkedIn Learning | Flutter Essential Training — LinkedIn Learning | Internet of Things

SOFT SKILLS & LANGUAGES

Problem Solving & Analytical Thinking | Team Collaboration & Adaptability | Stakeholder & Project Management | Time Management & Attention to Detail

Languages: English (Fluent), Malayalam (Fluent)